Western New England University
College of Engineering
ECE Department
RF & Microwave Active Circuit Design
EE 456/556
Spring 2025
Design Project #1
Due: February 24, 2025

Name: Steven Placzek

References: Ryan Leonard

Aidan Butler

Matt Saloio

Design Project #1

Element	Score	Max
Design		100
Input Matching Network		100
Output Matching Network		100
Simulations (MATLAB)		200
Simulations (ADS – Ideal TRLs)		100
Comparison		100
Presentation		100
Overall		800

Western New England University

EE 456

Design Project 1

by Steven Placzek

The objective of this project was to develop an amplifier that achieves maximum gain at 15 GHz using the MGF4941AL super-low noise InGaAs HeMT biased at V_DS = 2 V and I_DS = 10 mA.

NAME	TITLE	DWG. NO.
Steven Placek	EE456 - Design Project 1	IMN 1
SMITH CHART FORM ZY-01-N	COLOR BY J. COLVIN, UNIVERSITY OF FLORIDA, 1997	DATE 2/22/25

$\tilde{Z}_T = 1.0000$

$\tilde{y}_T = 1.0000$

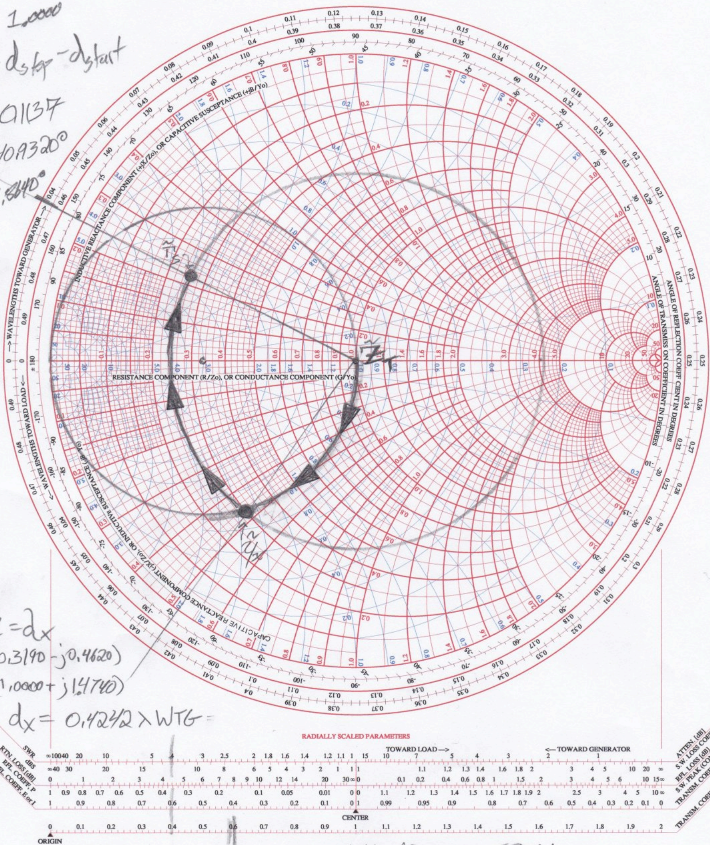
$d_{02} = d_{stop} - d_{start}$

$d_{02} = 0.1137$

$\theta_{02} = 40.9320^\circ$

$2\theta_{02} = 81.8640^\circ$

NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES



$d_{start} = d_x$

$Z_x = (0.3190 - j0.4620)$

$y_x = (1.0000 + j1.4740)$

$d_x = 0.4242\lambda$ WTG =

VSWR = 0.5871

$\Gamma_L = (0.5871 \angle +152.4980^\circ)$

NAME: Steven Pruczek

TITLE: EE456 - Design Project 1

DWG. NO.: IMN 2

DATE: 2/22/25

$d_{11} = d_{stop} - d_{start}$

$d_{stop} = d_{stub}$

$d_{stub} = 0.0759$ WTG

$\tilde{y}_x = 1.0000 - j1.9700$

$\tilde{y}_{stub} = \tilde{y}_x - \tilde{y}_T$

$\tilde{y}_{stub} = -j1.9700$

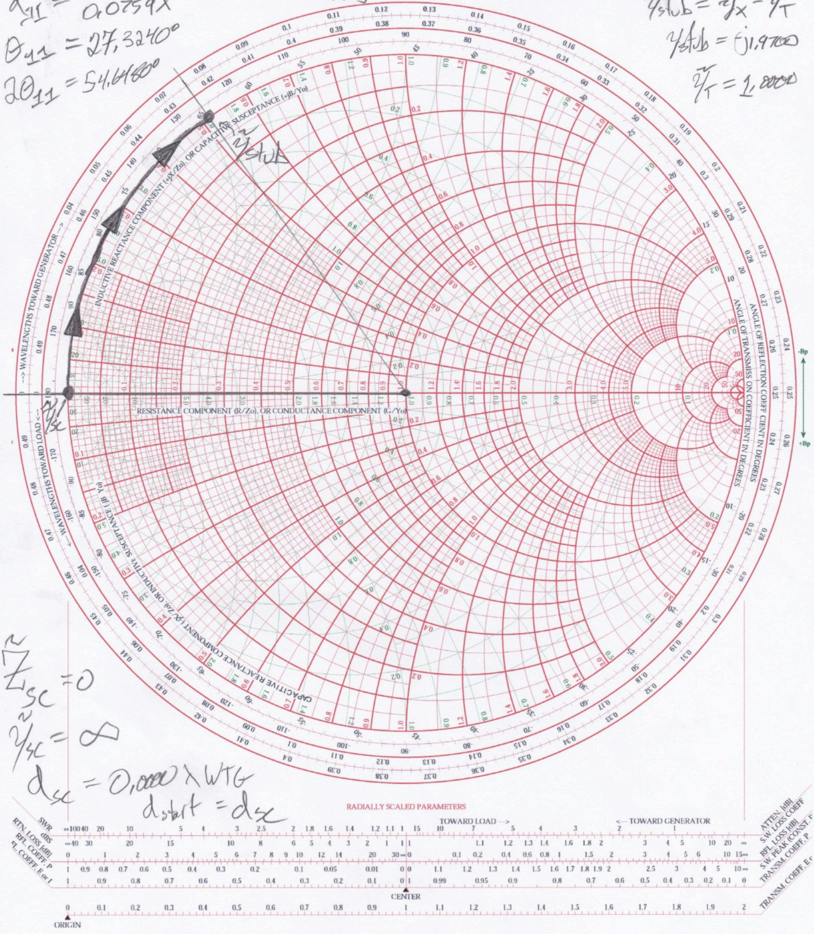
$\tilde{y}_T = 1.0000$

$d_{11} = 0.0759\lambda$

$\theta_{11} = 27.3240^\circ$

$2\theta_{11} = 54.6480^\circ$

NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES



$\tilde{Z}_{sc} = 0$

$\tilde{y}_{sc} = \infty$

$d_{sc} = 0.0000\ \lambda$ WTG

$d_{start} = d_{sc}$

NAME	TITLE	DWG. NO.
Steven Placzek	EE 456 - Design Project 1	OMN 1
		DATE 2/22/25

$d_{12} = d_{stop} - d_{start}$

$d_{12} = 0.3470\,\lambda$ WTG

$\theta_{12} = 124.9200^\circ$

$2\theta_{12} = 249.8400^\circ$

$Z_T = 1.0000 \quad Y_T = 1.0000$

NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES

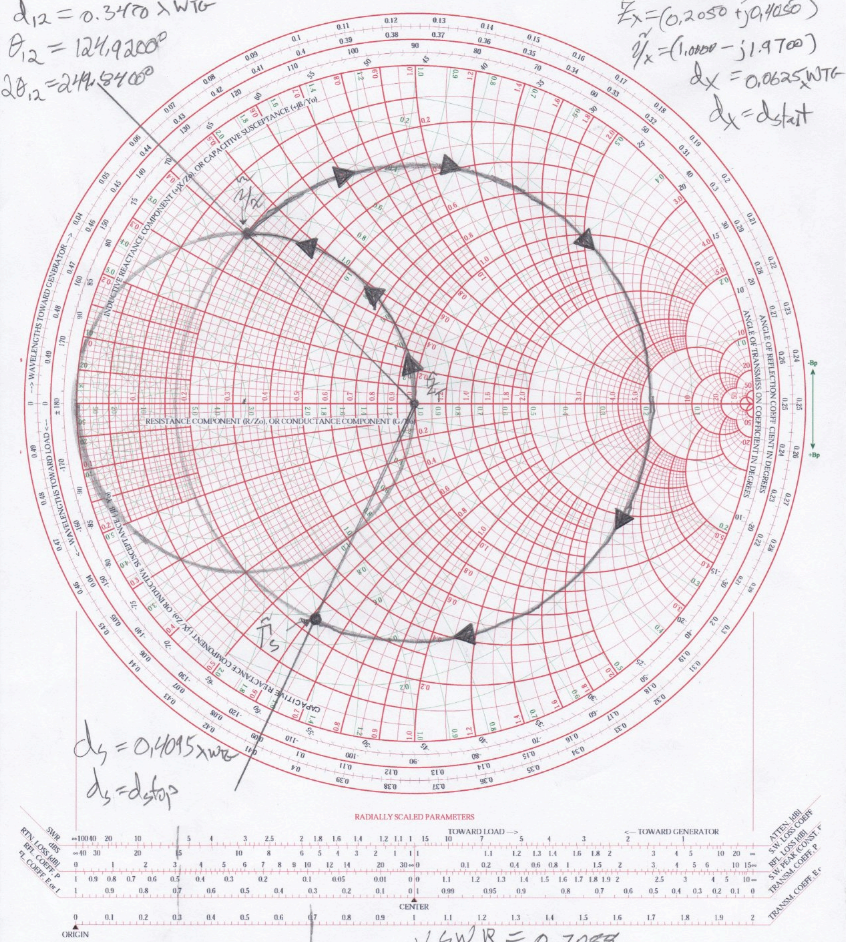
$\Gamma_S = (0.7088 \angle -114.5580^\circ)$

$\tilde{Z}_x = (0.2050 + j0.4050)$

$\tilde{Y}_x = (1.0000 - j1.9700)$

$d_x = 0.0625\,\lambda$ WTG

$d_x = d_{start}$



$d_S = 0.4095\,\lambda$ WTG

$d_S = d_{stop}$

VSWR = 0.7088

NAME	TITLE	DWG. NO.
Steven Placzek	EE456 - Design Project 1	OMN 2
SMITH CHART FORM ZY-01-N	COLOR BY J. COLVIN, UNIVERSITY OF FLORIDA, 1997	DATE 2/22/25

$d_{stop} = d_{stub}$ $\tilde{y}_T = 1.0000$

$d_{stub} = 0.4045\lambda$ WTG

$d_{01} = 0.1545$

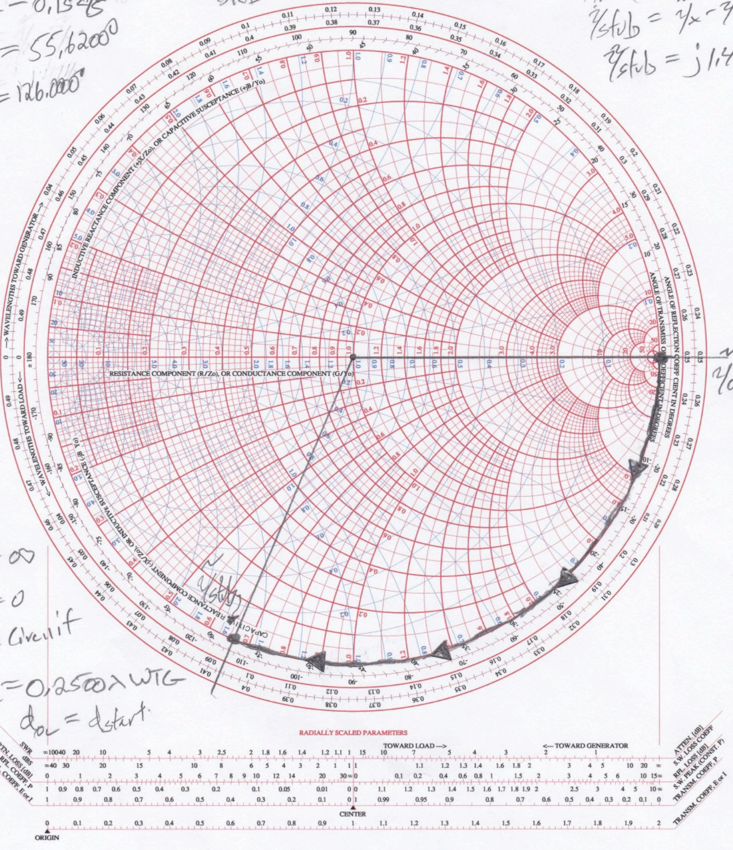
$\theta_{01} = 55.6200°$

$2\theta_{01} = 126.0000°$

$\tilde{y}_x = (1.0000 + j1.4740$

$\tilde{y}_{stub} = \tilde{y}_x - \tilde{y}_T$

$\tilde{y}_{stub} = j1.4740$



$\tilde{Z}_{oc} = \infty$

$\tilde{y}_{oc} = 0$

Open Circuit

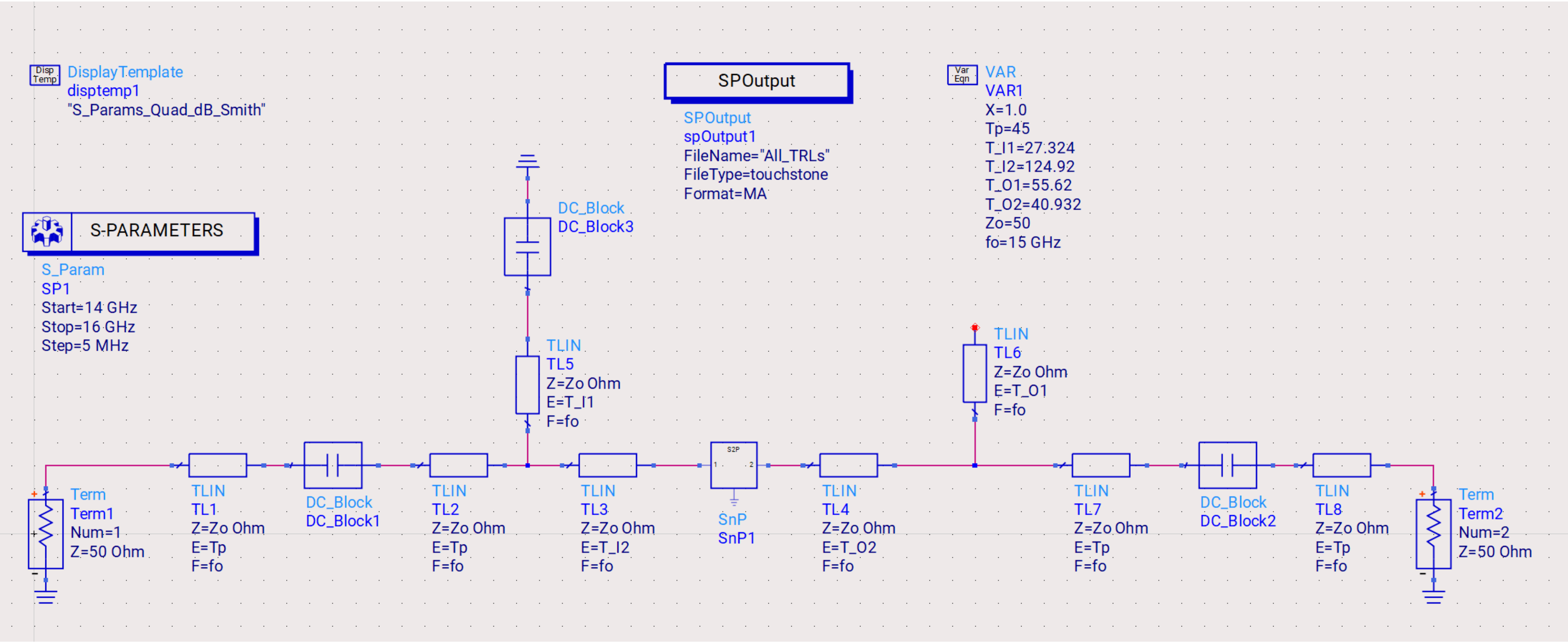
$d_{oc} = 0.2500\lambda$ WTG

$d_{oc} = d_{start}$

Complete Design Schematic

Amplifier Designed for Maximum Available Gain

— created using KEYSIGHT ADS SOFTWARE

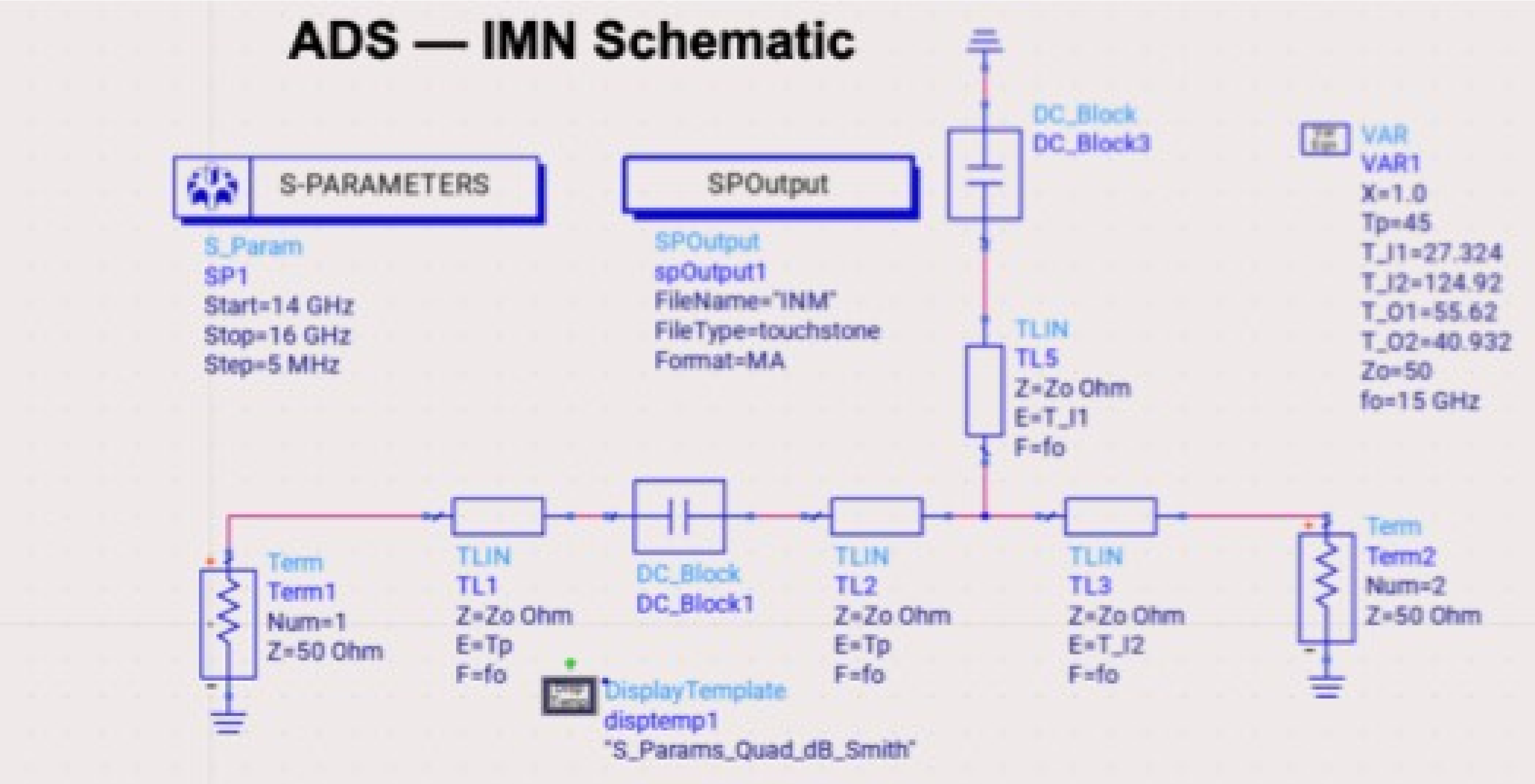


*Schematic above includes every Transmission Line and matching network within the Project Scope**

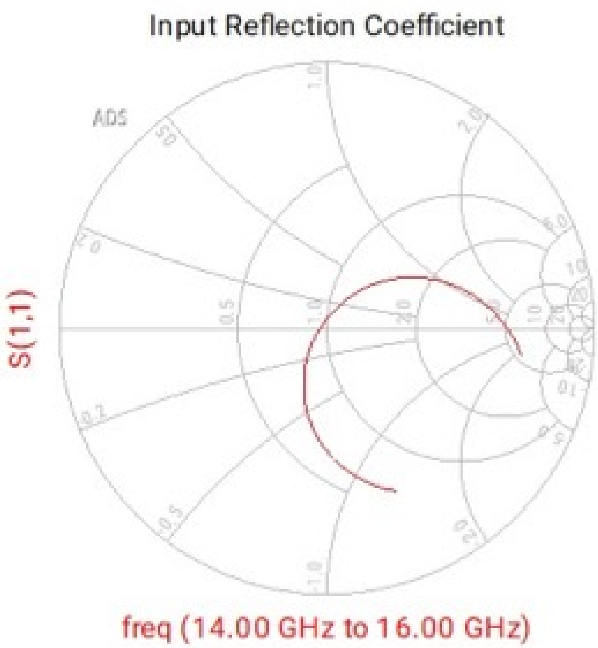
*(*All Simulations exclusively employ an Ideal Transmission Line model for Parameter Values)*

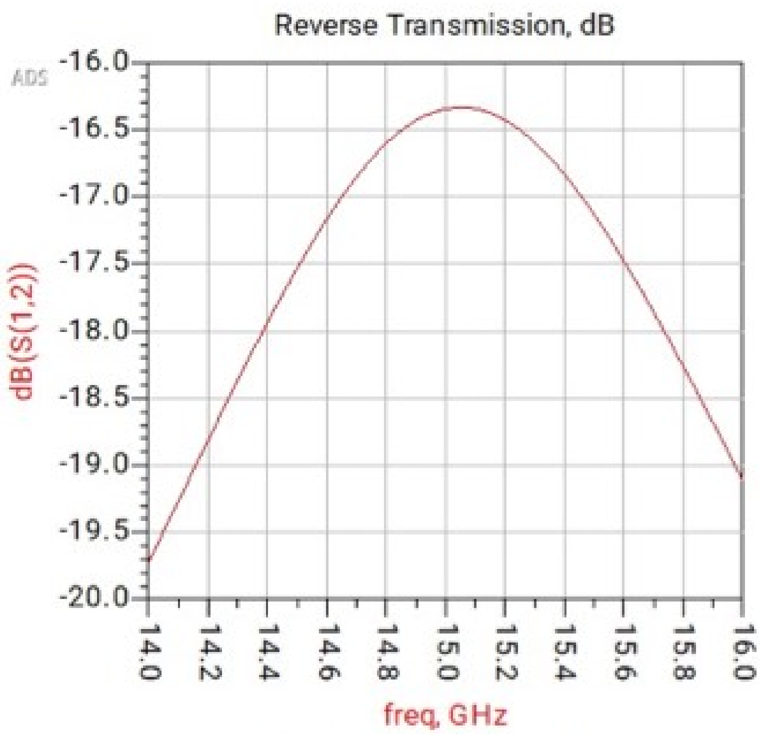
Keysight ADS Simulations Results for the IMN

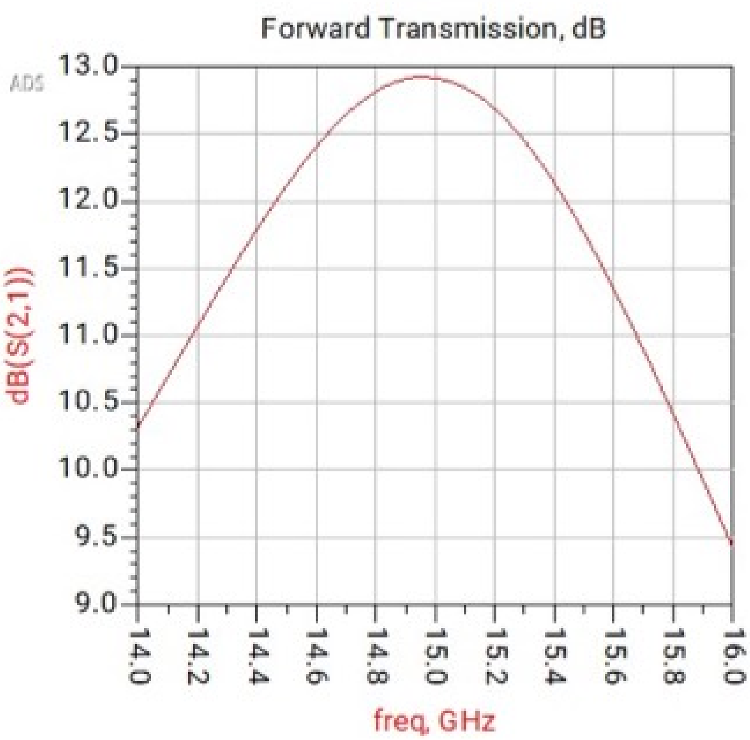
Input Matching Network Design: Smith Charts & S Parameters

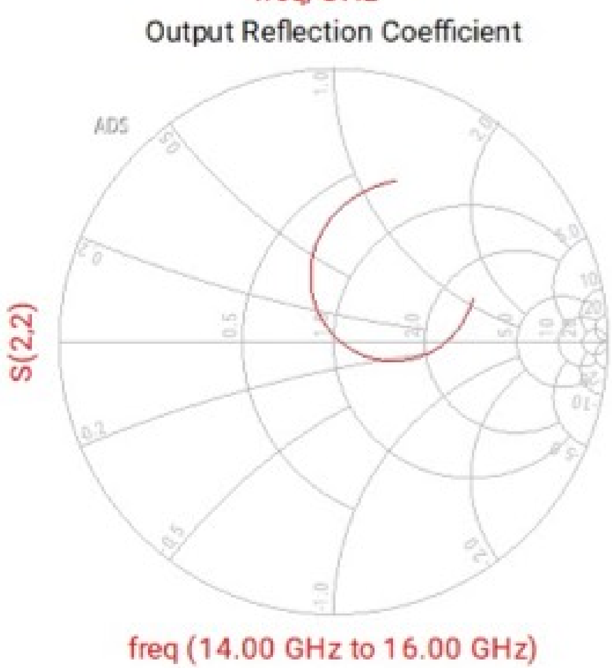


IMN Design Simulation Results: Smith Charts & S Parameters

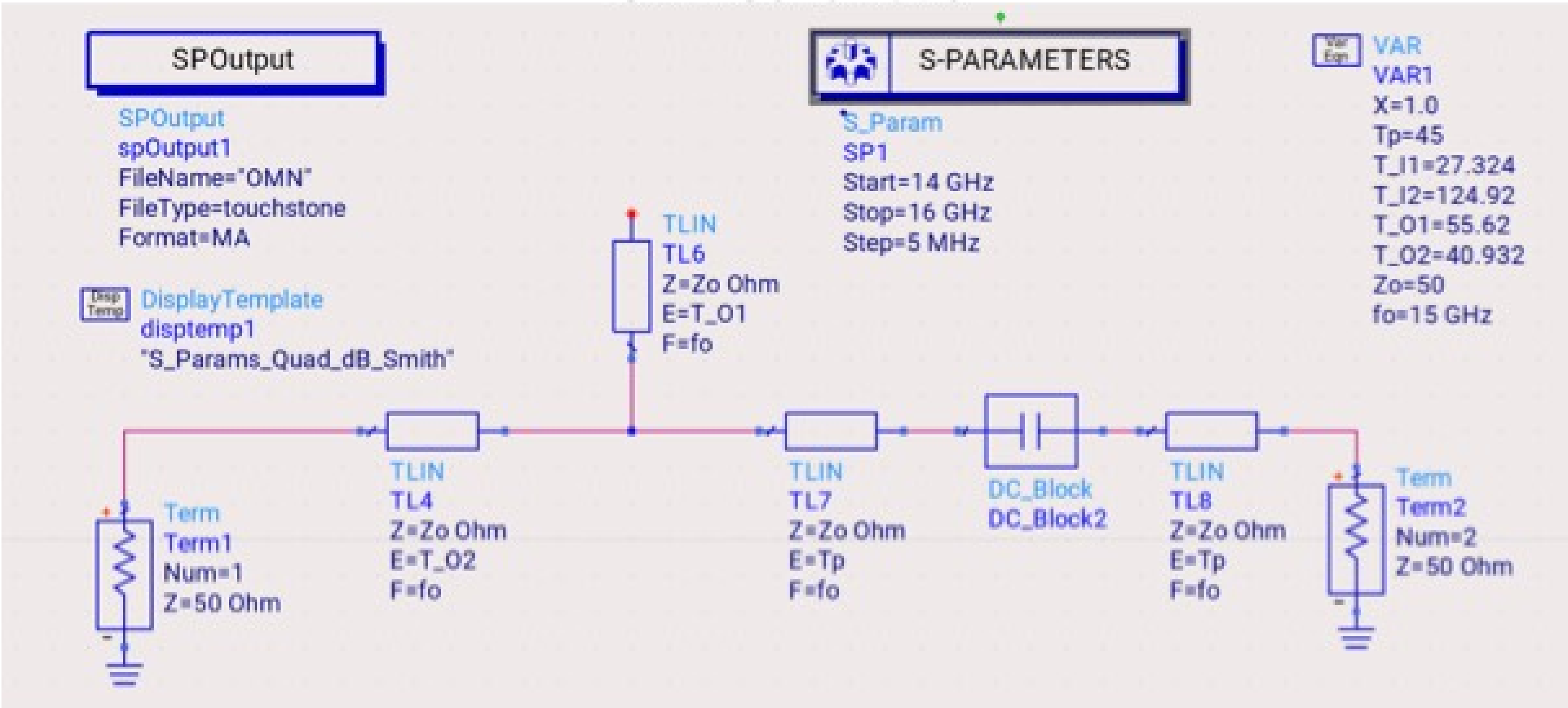




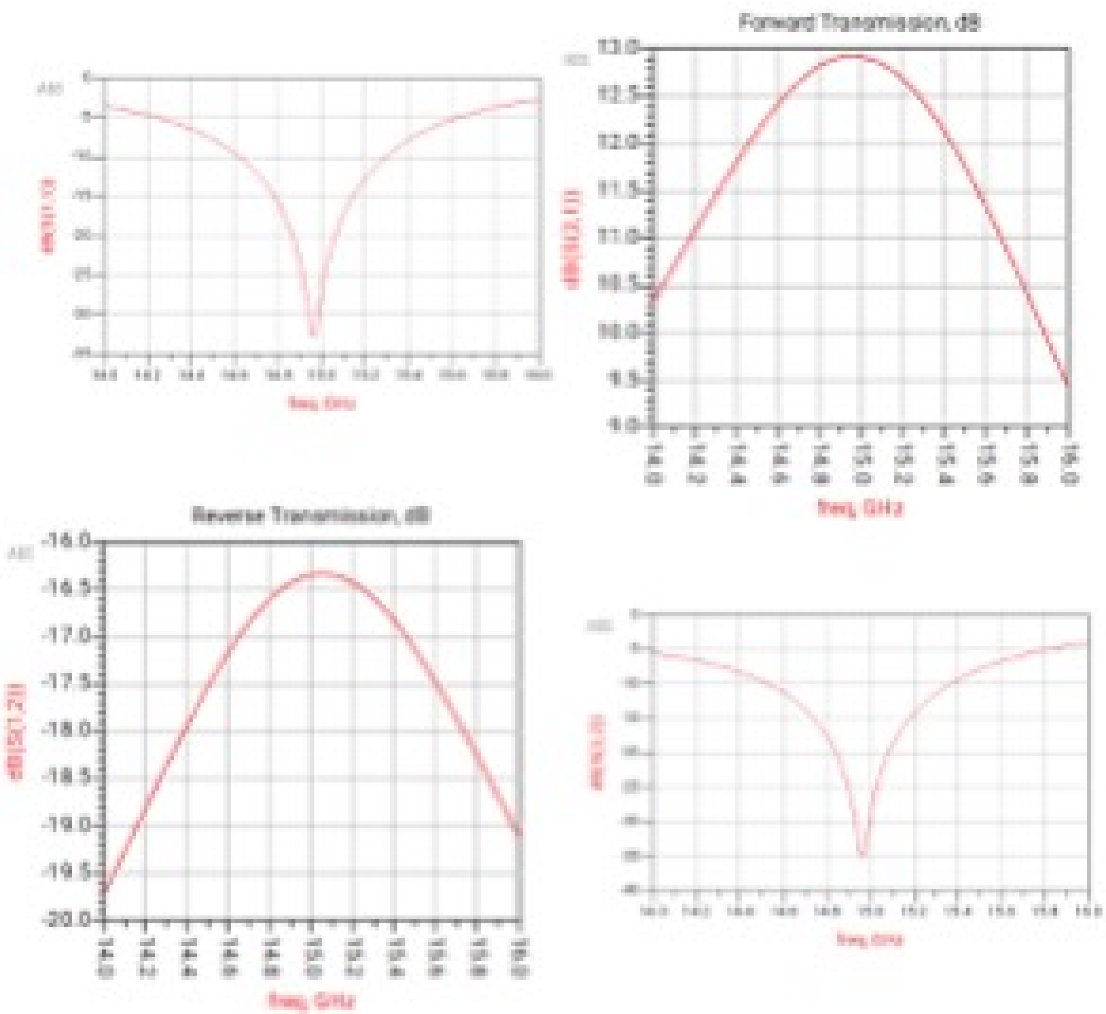




OMN Schematic

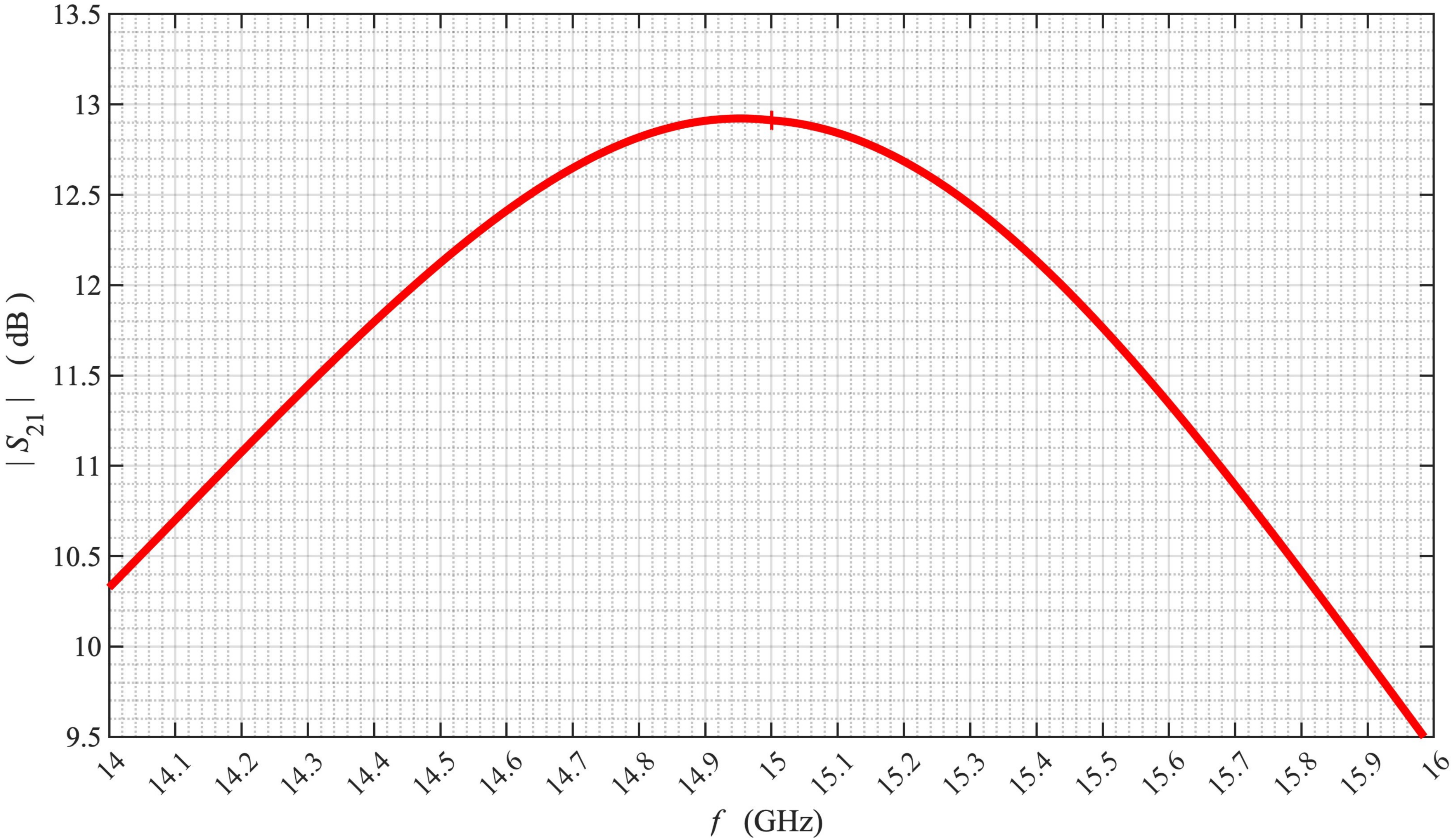


ADS — S Parameters vs Frequency Plots



Task 4 — Part 1 — Amplifier Design Parameter Simulation via Matlab

|S21| Gain (dB) vs Frequency (Hz) Plot

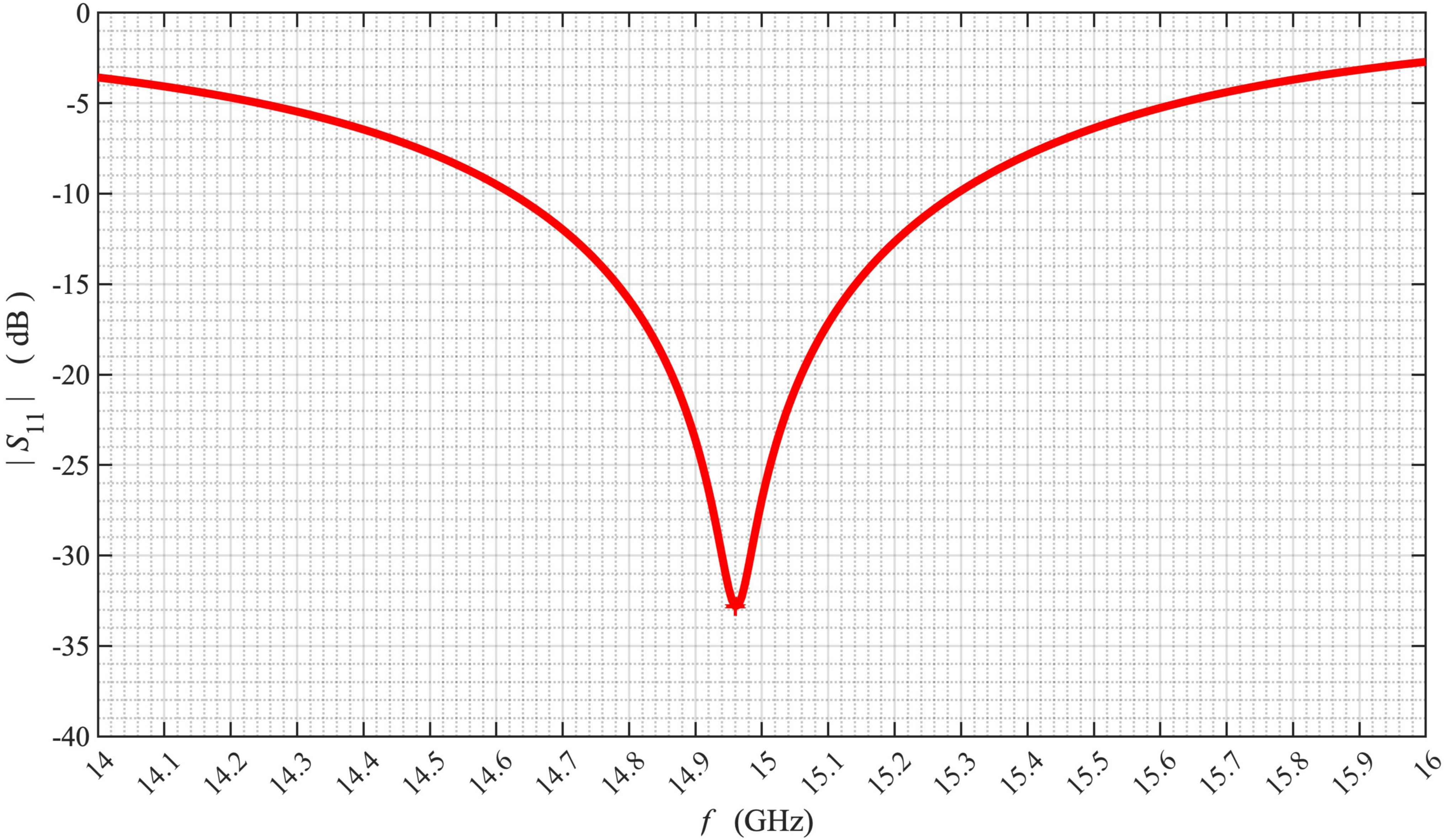


(Plot generated using Matlab)

[Simulations exclusively employ Ideal Transmission Lines]

Task 4 — Part 2 — Simulate Amplifier Design Parameters via Matlab

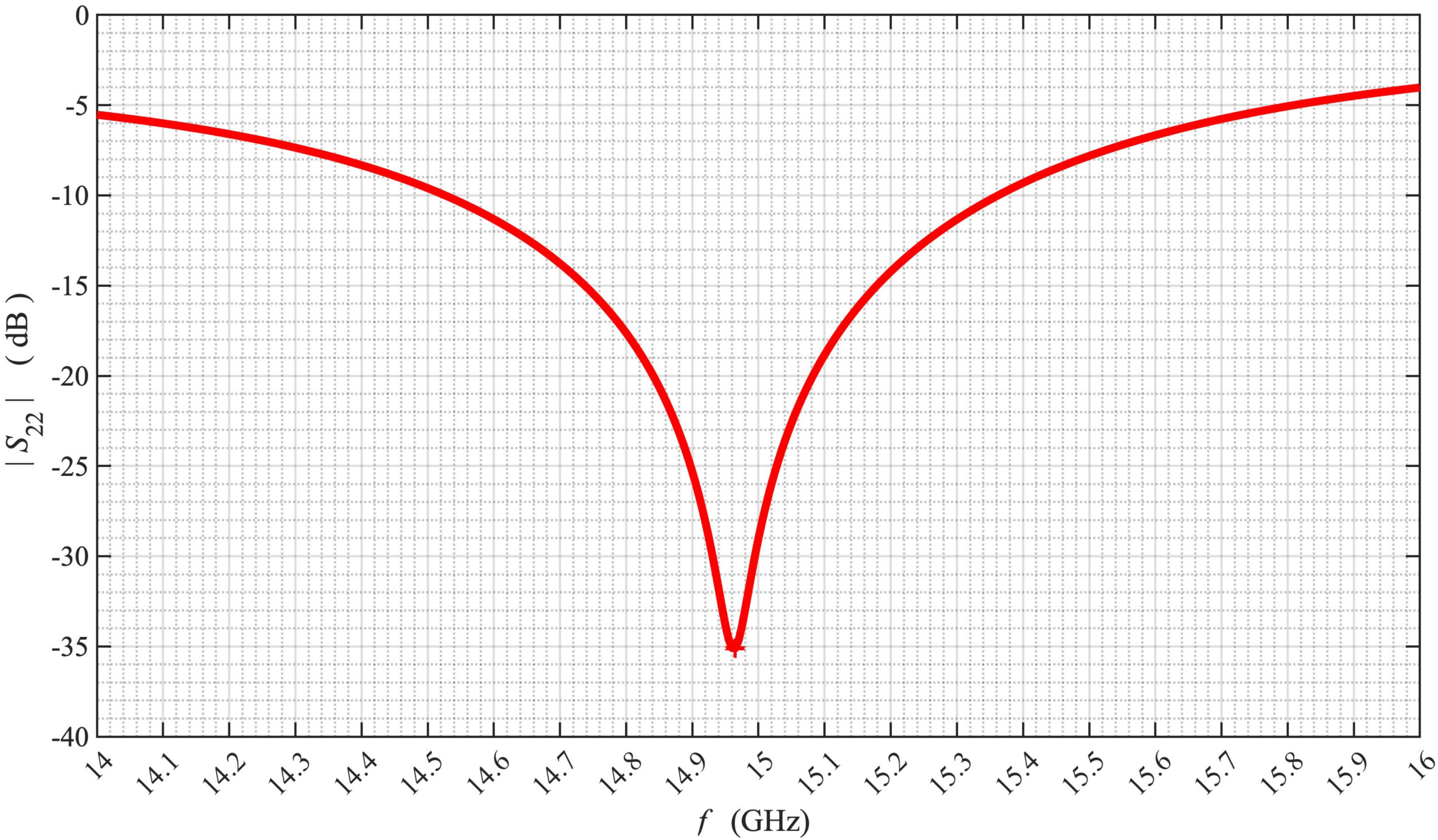
|S11| Gain (dB) vs Frequency (Hz) Plot



(Plot generated using Matlab)

Task 4 — Part 3 — Simulate Amplifier Design Parameters via Matlab

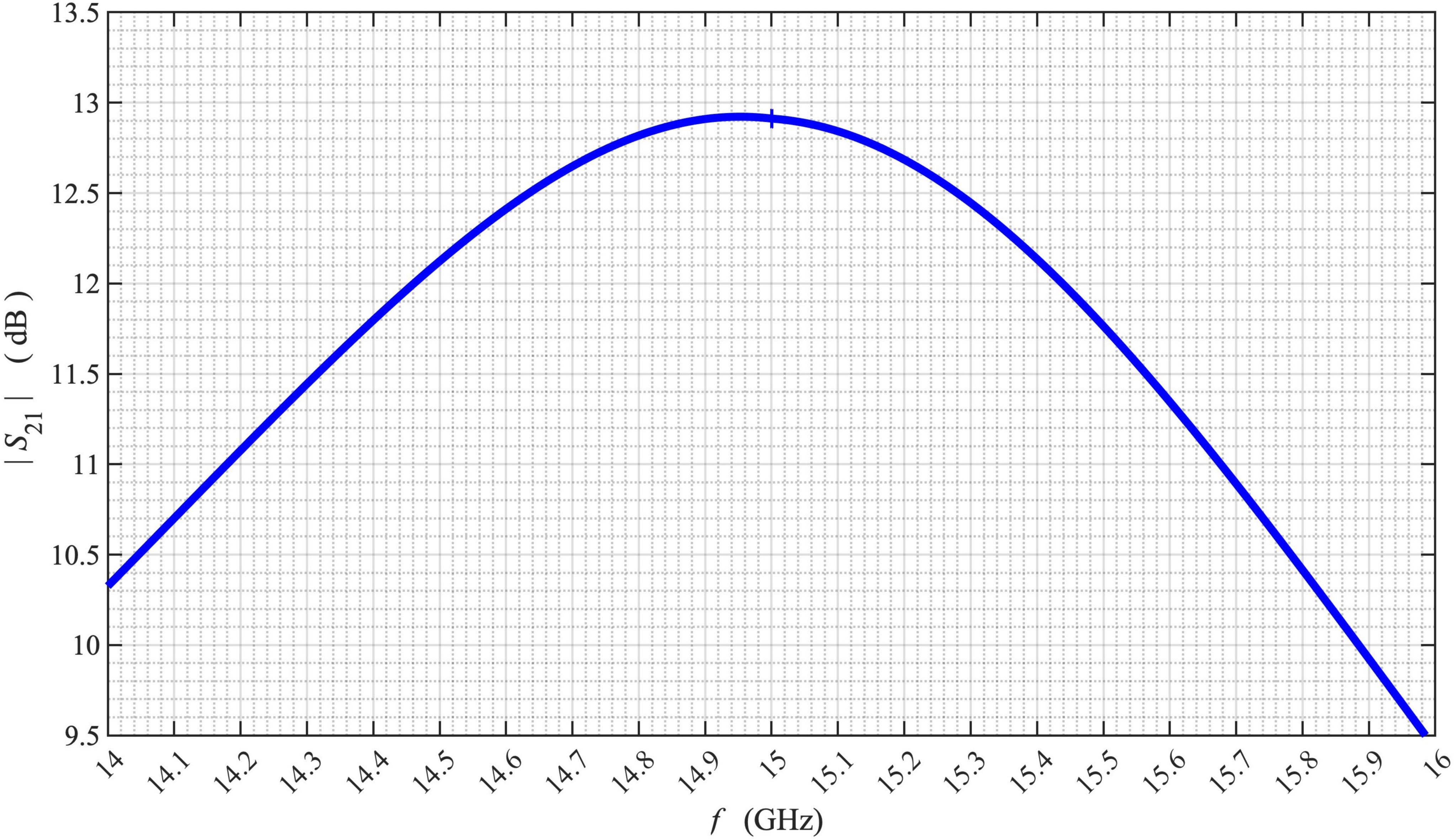
|S22| Gain (dB) vs Frequency (Hz) Plot



(Plot generated using Matlab)

Task 5 — Part 1 — Plotting Keysight ADS Results with Matlab

|S21| Gain (dB) vs Frequency (Hz) Plot

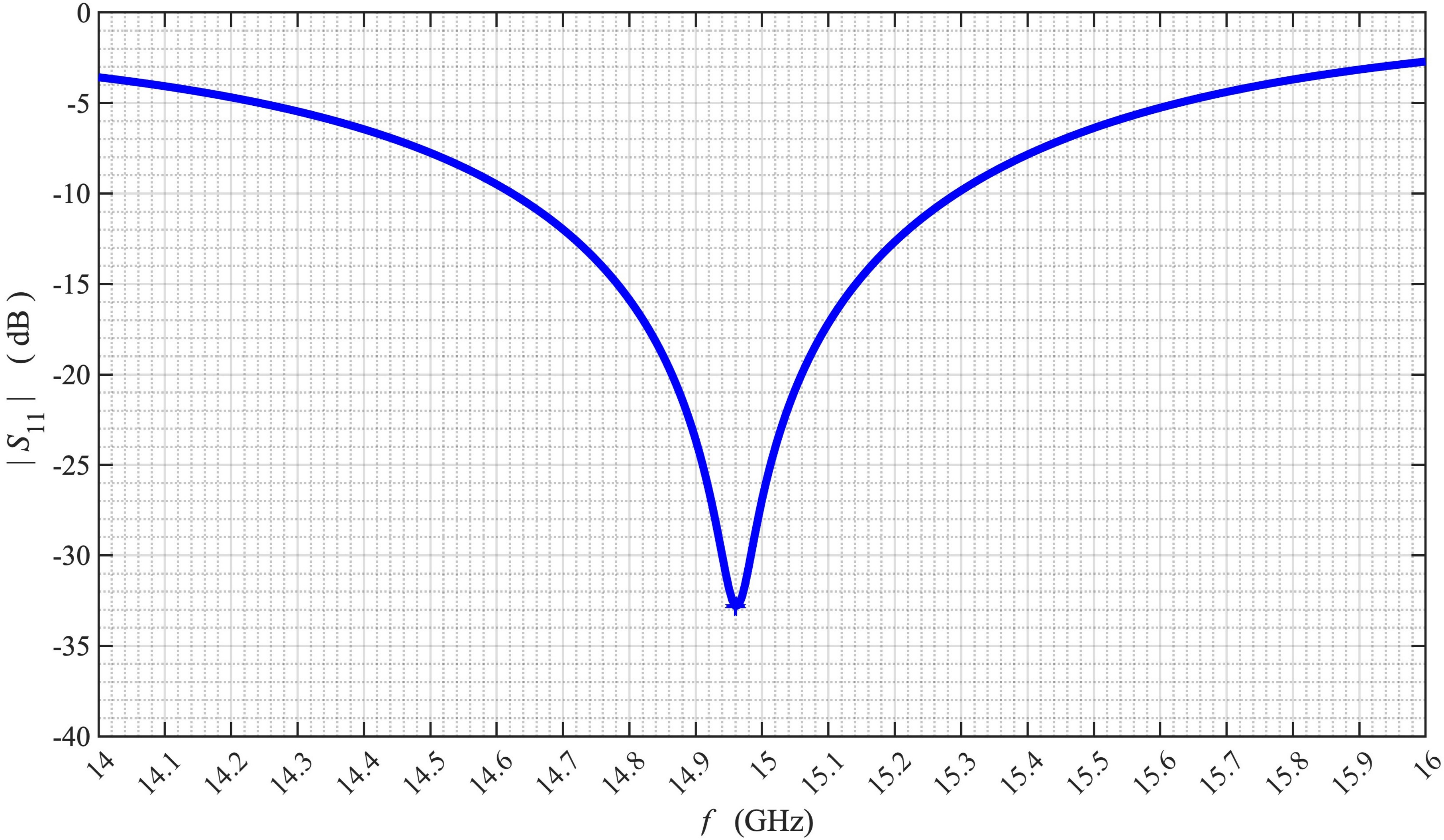


[Simulations exclusively employ Ideal Transmission Lines]

(Plot generated using Matlab)

Task 5 — Part 2 — Keysight ADS Simulation Results

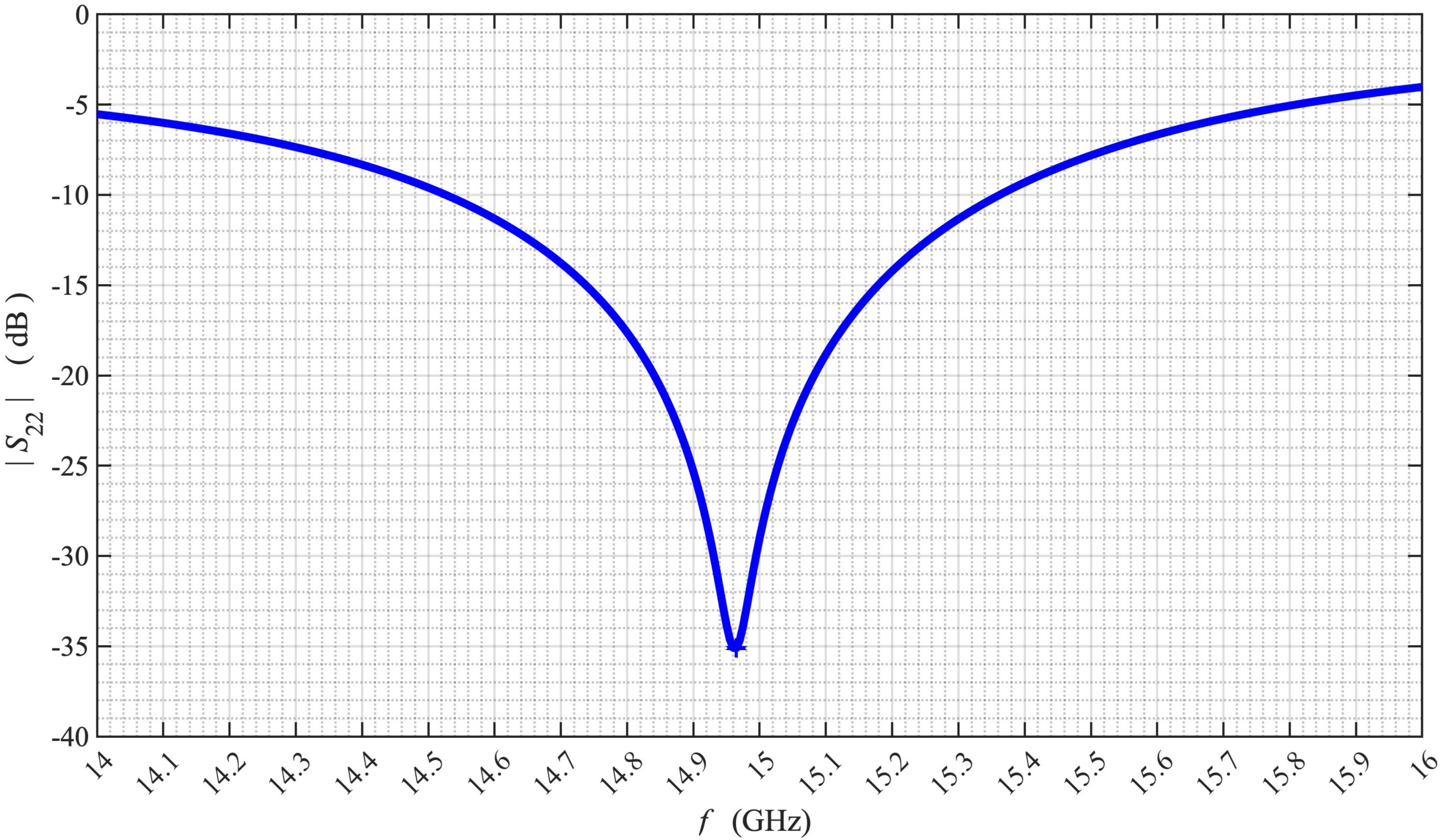
|S11| Gain (dB) vs Frequency (Hz) Plot



[Simulations exclusively employ Ideal Transmission Lines]

(Plot generated using Matlab)

Task 5 — Part 3 — Plotting Keysight ADS Results with Matlab
|S22| Gain (dB) vs Frequency (Hz) Plot

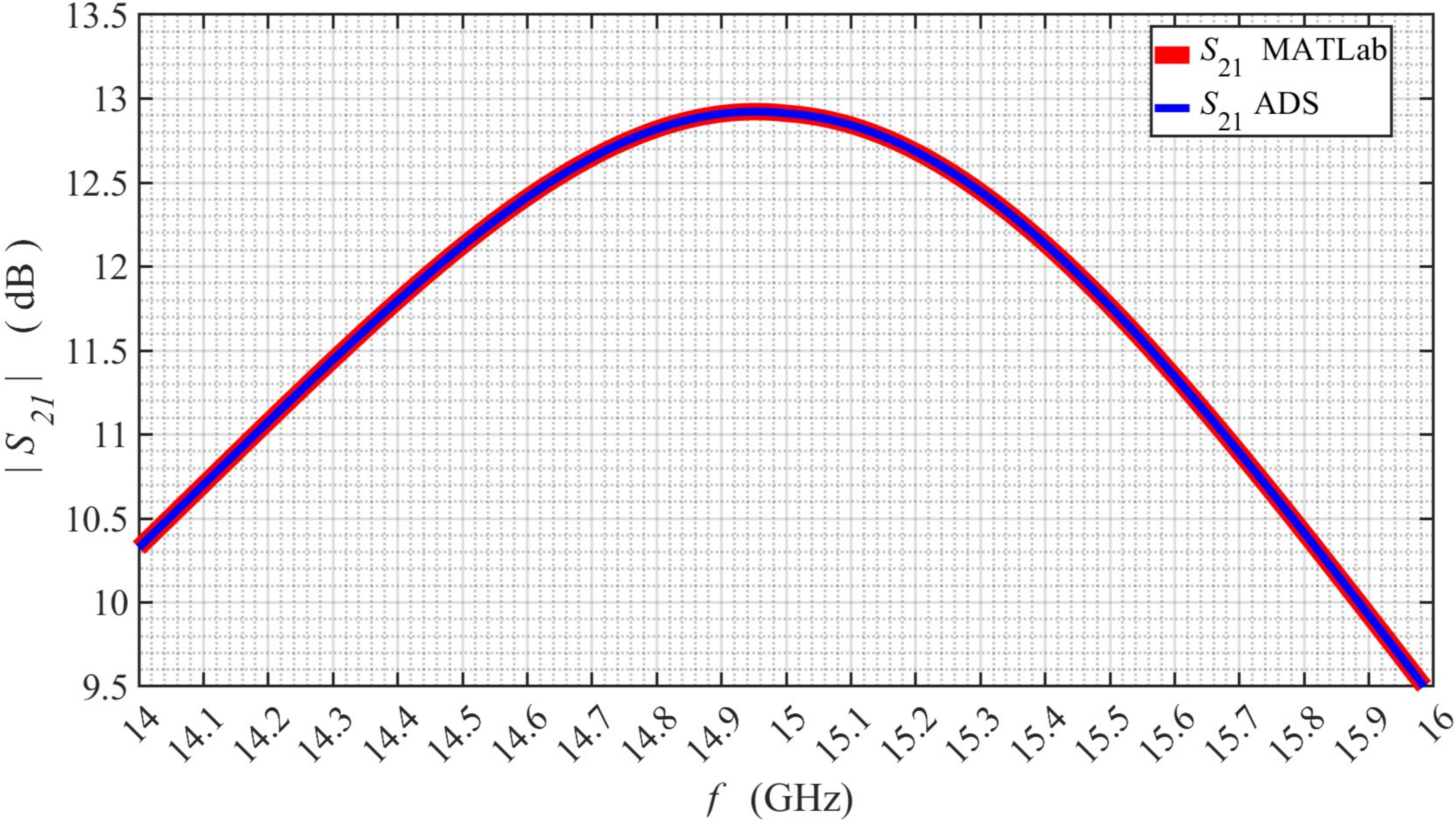


(All Simulations exclusively employ Ideal Transmission Lines)

(*Plot generated using Matlab*)

Task 6 — Part 2 — Matlab vs ADS Simulation Comparisons

|S21| Gain (dB) vs Frequency (Hz) Plot

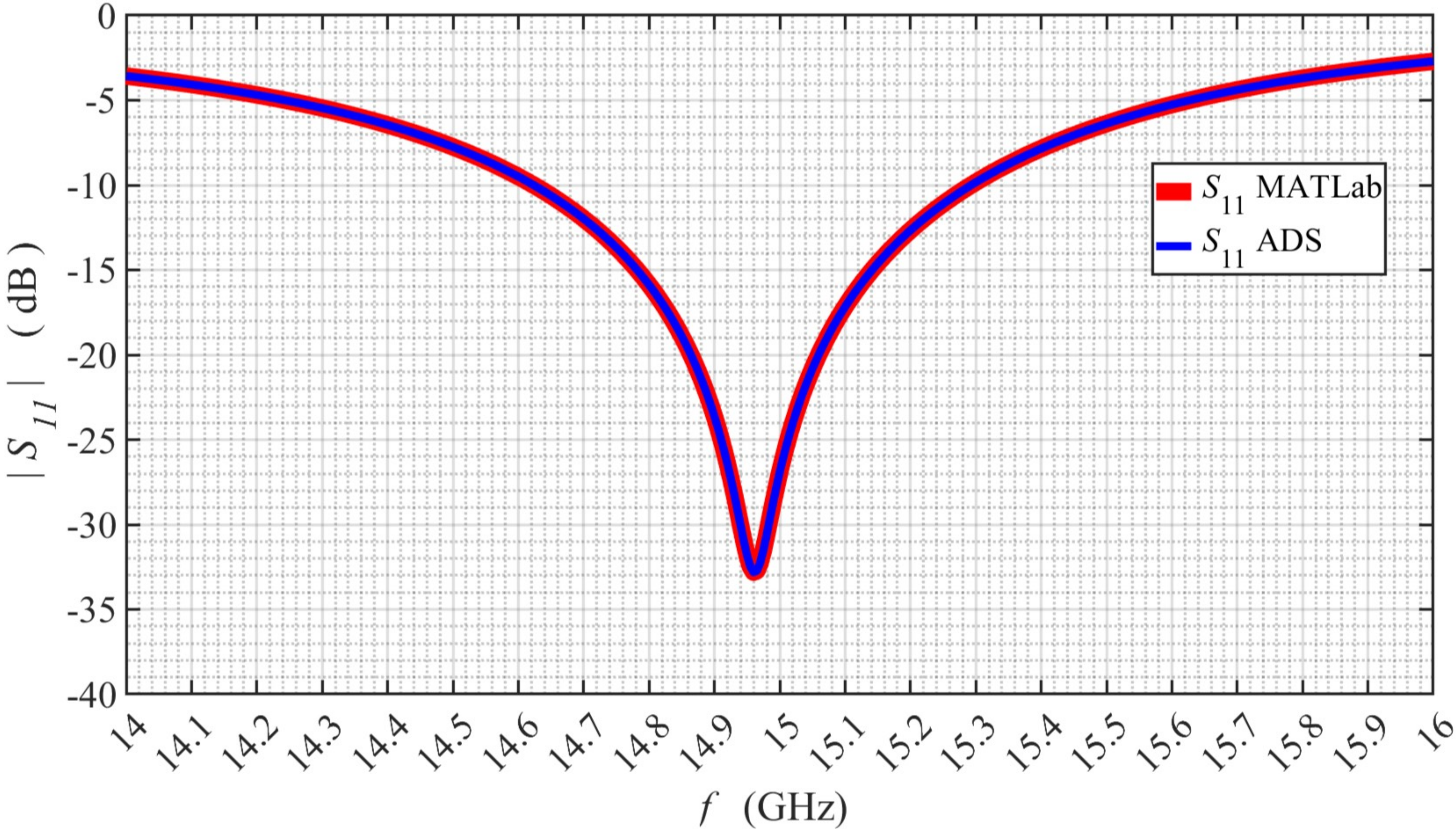


(All Simulations exclusively employ Ideal Transmission Lines)

(*Plot generated using Matlab*)

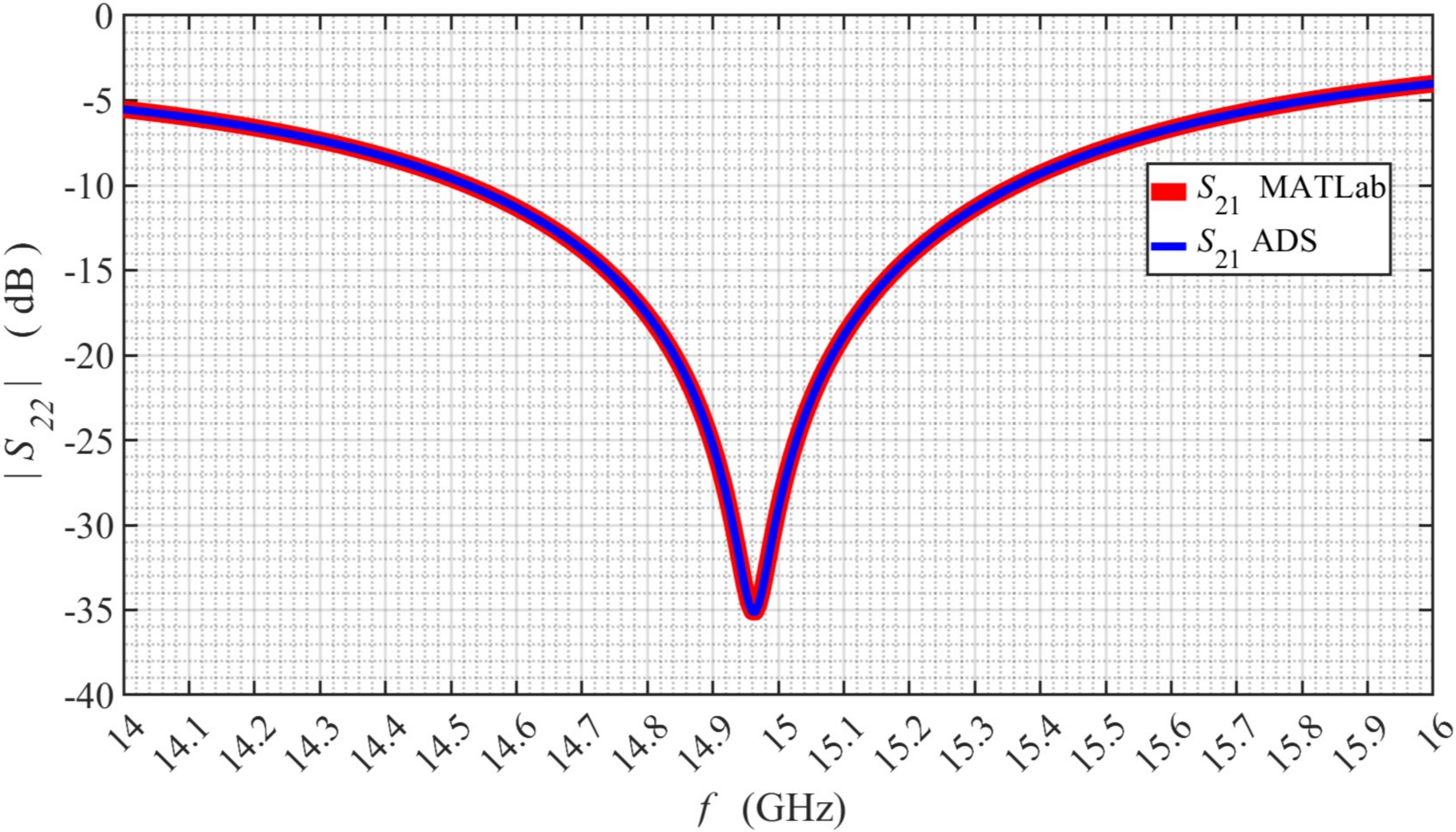
Task 6 — Part 2 — Matlab vs ADS Simulation Comparisons

|S11| Gain (dB) vs Frequency (Hz) Plot



(*Plot generated using Matlab*)

Task 6 — Part 3 — Matlab vs ADS — Simulation Comparisons
|S22| Gain (dB) vs Frequency (Hz) Plot



(*Plot generated using Matlab*)

Western New England University

EE 456

Design Project 1

Task 7: *Complete Table 1*

Table 1:

Parameter	Design	Simulated_MATLAB	Simulated_ADS
Γ_s	0.7088∠-114.5883°	0.6954∠-23.9007°	0.6910∠-113.3649°
Γ_L	0.5871∠152.4980°	0.5900∠151.9775°	0.5218∠162.0738°
Γ_in	0.7088∠114.5883°	0.7104∠114.3690°	0.3593∠158.3169°
Γ_out	0.5871∠-152.4980°	0.2614∠-75.0200°	0.4488∠-112.3242°
|Γ_IMN|	0.0000	0.0000	0.5994
|Γ_IMN| (dB)	-307.0086	-0.9925	-4.4461
VSWR_IMN	1.0000	17.5223	14.8120
|Γ_OMN|	0.0000	0.6029	0.4769
|Γ_OMN| (dB)	-307.4626	-4.3950	-6.4319
VSWR_OMN	1.0000	4.0366	3.7192
GT (W/W)	19.5936	19.5936	19.5527
GT (dB)	12.9211	12.9211	12.9121

EE456_Design_Project_1_Placzek_Task_7_Table_1

Parameter	Design	Simulated_MATLAB	Simulated_ADS
Γ_s	0.7088∠-114.5883°	0.6954∠-23.9007°	0.6910∠-113.3649°
Γ_L	0.5871∠152.4980°	0.5900∠151.9775°	0.5218∠162.0738°
Γ_in	0.7088∠114.5883°	0.7104∠114.3690°	0.3593∠158.3169°
Γ_out	0.5871∠-152.4980°	0.2614∠-75.0200°	0.4488∠-112.3242°
|Γ_IMN|	0.0000	0.0000	0.5994
|Γ_IMN| (dB)	-307.0086	-0.9925	-4.4461
VSWR_IMN	1.0000	17.5223	14.8120
|Γ_OMN|	0.0000	0.6029	0.4769
|Γ_OMN| (dB)	-307.4626	-4.3950	-6.4319
VSWR_OMN	1.0000	4.0366	3.7192
GT (W/W)	19.5936	19.5936	19.5527
GT (dB)	12.9211	12.9211	12.9121

Results / Analysis

— This design demonstrates excellent matching for this project, with both VSWR_IMN and VSWR_OMN being nearly 1. Additionally, Gamma_OMN and Gamma_IMN differ by less than 0.05.

— The overall gain, G_T, is very on target toward achieving the theoretical maximum, indicating an exact match.